

KONFIGURASI VPN SERVER DI LINUX DEBIAN 11

Virtual Private Network salah satu teknologi yang diterapkan di internet untuk koneksi aman terenkripsi dimana koneksi tersebut bisa di akses tanpa harus menuju ke destination (atau bisa dikatakan maya/tidak terlihat).

Untuk melakukan tahapan install VPN pada debian pastikan sudah masuk pada terminal linux menggunakan login user root agar semua akses bisa dijalankan.

1. Setting adapter 1 debian dengan NAT, sedangkan untuk adapter 2 = internal network
2. Pastikan debian mendapatkan IP = 10.0.2.15, agar debian anda dapat mengakses internet.
3. Konfigurasikan IP untuk adapter 2 (sesuai perintah soal)
4. Install layanan untuk VPN : **apt-get install pptpd -y**
5. Masuk ke file konfigurasi VPN : **nano /etc/pptpd.conf**

```
GNU nano 5.4 /etc/pptpd.conf
# 1. No spaces are permitted between commas or within addresses.
#
# 2. If you give more IP addresses than the value of connections,
#    it will start at the beginning of the list and go until it
#    gets connections IPs. Others will be ignored.
#
# 3. No shortcuts in ranges! ie. 234-8 does not mean 234 to 238,
#    you must type 234-238 if you mean this.
#
# 4. If you give a single localIP, that's ok - all local IPs will
#    be set to the given one. You MUST still give at least one remote
#    IP for each simultaneous client.
#
# (Recommended)
localip 10.10.10.1
remoteip 10.10.10.2-10
# or
#localip 192.168.0.234-238,192.168.0.245
#remoteip 192.168.1.234-238,192.168.1.245
```

Edit pada bagian yang di blok (sesuaikan dengan soal/ localip adalah IP VPN yang akan digunakan)

6. Masuk ke file untuk konfigurasi dns : **nano /etc/ppp/pptpd-options**

```
GNU nano 5.4 /etc/ppp/pptpd-options
#chapms-strip-domain

# Encryption
# (There have been multiple versions of PPP with encryption support,
# choose with of the following sections you will use.)

# BSD licensed ppp-2.4.2 upstream with MPPE only, kernel module ppp_mppe.
# [[[
#refuse-pap
#refuse-chap
#refuse-mschap
# Require the peer to authenticate itself using MS-CHAPv2 [Microsoft
# Challenge Handshake Authentication Protocol, Version 2] authentication.

GNU nano 5.4 /etc/ppp/pptpd-options
# option allows pptpd to supply one or two DNS (Domain Name Server)
# addresses to the clients. The first instance of this option
# specifies the primary DNS address; the second instance (if given)
# specifies the secondary DNS address.
# Attention! This information may not be taken into account by a Windows
# client. See KB311218 in Microsoft's knowledge base for more information.
ms-dns 8.8.8.8
ms-dns 8.8.4.4
```

Edit pada bagian yang di blok.

7. Kemudian masuk ke file chap-secrets : **nano /etc/ppp/chap-secrets**

```
GNU nano 2.7.4 File: /etc/ppp/chap-secrets

# Secrets for authentication using CHAP
# client      server  secret          IP addresses
tori         pptpd   123456          *
```

8. Restart layanan pptpd : **sudo service pptpd restart**

9. Aktikan file rc.local

nano /etc/rc.local

isi dengan :

```
#!/bin/sh -e
#
# rc.local
exit 0
```

Kemudian simpan file nya.

Tulis perintah seperti di bawah ini :

chmod +x /etc/rc.local

systemctl daemon-reload

systemctl start rc-local

systemctl status rc-local → pastikan statusnya aktif

10. Share internet dari server debian virtual ke client windows virtual

nano /etc/sysctl.conf

```
GNU nano 5.4 /etc/sysctl.conf
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1

# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1
```

nano /etc/rc.local

```
GNU nano 5.4 /etc/rc.local
#!/bin/sh -e
#
#rc.local

iptables -t nat -A POSTROUTING -o enp0s3 -j MASQUERADE

exit 0
```

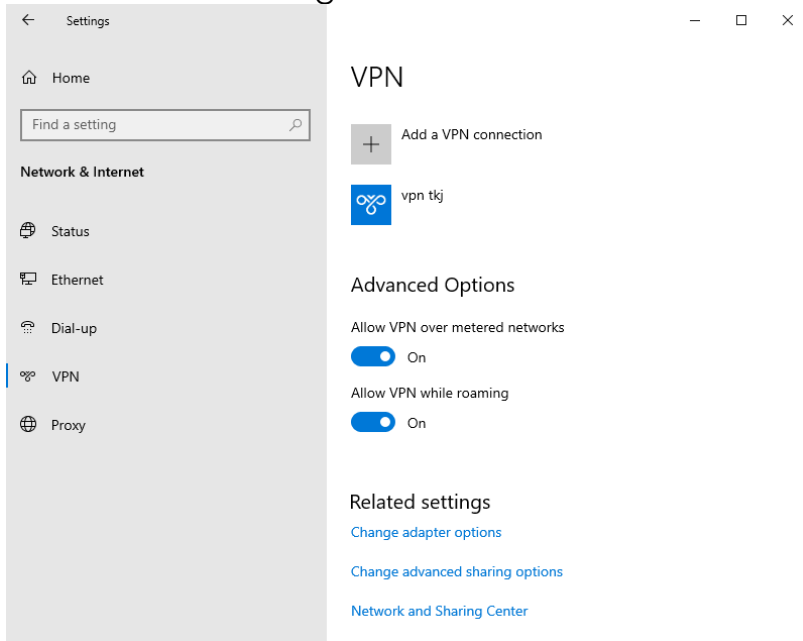
reboot

Pastikan windows virtual/client dapat akses internet

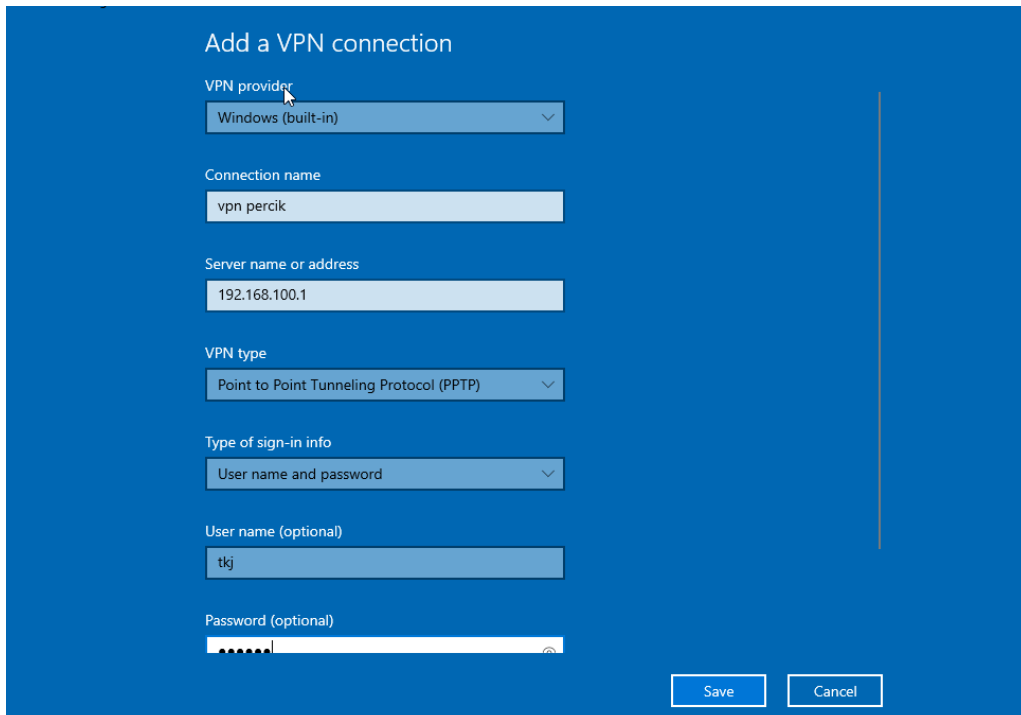
11. Konfigurasi VPN Client

Isi IP client dengan IP yang satu jaringan dengan server, kemudian cek konektivitas dari client ke server, begitu juga sebaliknya.

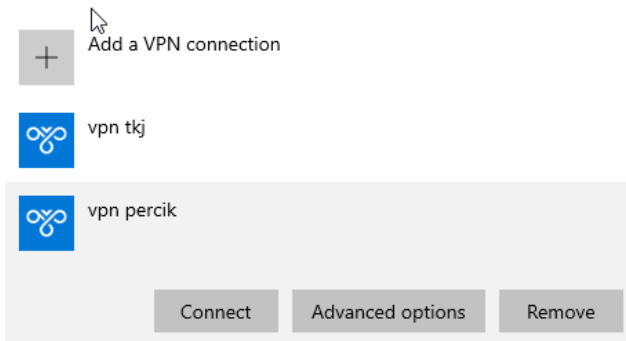
Masuk ke VPN setting



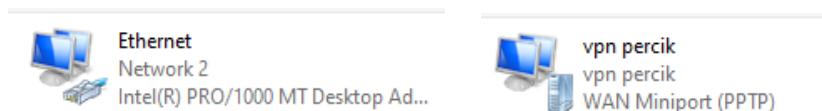
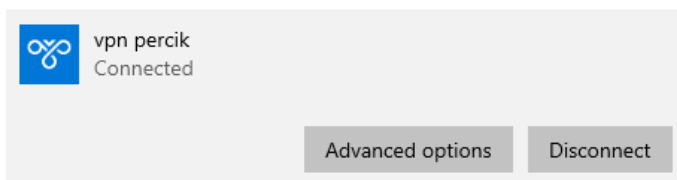
kemudian Add a VPN connection



VPN



Jika koneksi VPN berhasil, maka akan muncul seperti gambar di bawah ini :



Pada command prompt client, ketikkan perintah `tracert google.com`, pastikan hasilnya terdapat local ip yang di server debian.

```
Command Prompt
Microsoft Windows [Version 10.0.19045.2965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MasterWindows>tracert google.com

Tracing route to forcesafesearch.google.com [216.239.38.120]
over a maximum of 30 hops:

  1      1 ms    <1 ms    <1 ms    10.10.10.1
  2     16 ms    15 ms    15 ms    any-in-2678.1e100.net [216.239.38.120]

Trace complete.

C:\Users\MasterWindows>
```